

Chapter 4

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Claim 4.1. *A space $(X, \mathcal{T})$ is T_1 if and only if every point of X is a closed set*

Proof. $\implies$: Suppose $(X, \mathcal{T})$ is T_1 . By definition of T_1 space, $\forall x, y \in X \ni x \neq y, \exists V \in \mathcal{T} \ni y \in V$ but $x \notin V$. Thus y is not a limit point of $\{x\}$ if $y \neq x$. Thus the closure of $\{x\}$ is $\{x\}$. Thus each point is a closed set.

$\impliedby$: Suppose every point of X is a closed set. Then by the negation of the definition of a limit point $\forall x, y \in X \ni x \neq y, \exists U, V \in \mathcal{T} \ni x \in U, y \in V, x \notin V, y \notin U$. Thus by the definition of T_1 spaces, $(X, \mathcal{T})$ is a T_1 space. $\square$

Problem 4.2. *Let X be a space with a finite complement topology. Show X is T_1*

Proof. $\forall x, y \in X \ni x \neq y, \exists U, V \in \mathcal{T} \ni U = X \setminus \{y\}, V = X \setminus \{x\}$ thus $x \in U$ and $y \in V$ but $y \notin U$ and $x \notin V$. Thus X is T_1 $\square$

Problem 4.3. *Show $\mathbb{R}_{std}$ is Hausdorff, T_2 .*

Proof. $\forall x, y \in \mathbb{R} \ni x \neq y$. WLOG $x < y$. Let $U = (-\infty, \frac{x+y}{2})$ and $V = (\frac{x+y}{2}, \infty)$. Since $x < y \implies x < \frac{x+y}{2} < y$. Thus $x \in U$ and $y \in V$. Assume the contrary $\exists z \in U \cap V \implies z < \frac{x+y}{2}$ and $x > \frac{x+y}{2}$ which is a contradiction. Thus $\mathbb{R}_{std}$ is Hausdorff. $\square$

Problem 4.4. *Show H_{bub} is regular*

Proof. $\forall p = (x, y) \in H$ and $\forall A \subset X$ where A is closed and $p \notin A$. Thus $\exists U \in \mathcal{T} \ni U \cap A = \emptyset$. Thus $\exists r \in \mathbb{R}^+ \ni B(p, r) \subseteq U$ since $B(p, r)$ is a basis element. Let $\delta = \min(|p - q| \mid q \in A)/2$. Let $V = B(p, \min(r, \delta))$ and let $W = \bigcup_{(a_x, a_y) \in A} B((a_x, a_y), \min(\delta, a_y))$. $p \in V \subseteq U$ and thus $V \cap A = \emptyset$. $A \subseteq W$.

Assume the contrary, $\exists q \in V \cap W$. Thus $q \in V \implies |p - q| < \min(r, \delta) \leq \delta = 0.25 * \min(|p - o| \mid o \in A) \implies 3 * \delta \leq |q - o| \forall o \in A$ which contradicts the fact that since $q \in W, \exists o \in A \ni |q - o| < \delta < 3\delta$. Thus by contradiction, $V \cap W = \emptyset$.

Thus H_{bub} is regular. $\square$

Problem 4.5. *R_{LL} is normal*

Proof. $\forall A, B \in 2^{\mathbb{R}}$ such that A, B are closed and $A \cap B = \emptyset$. $\forall a \in A$, a is not a limit point of B otherwise $a \in B$ which contradicts disjoint, $\exists U_a \in \mathcal{T} \ni U_a \cap B = \emptyset$. WLOG assume $U_a = [a, a + \epsilon_a)$ since $[a, a + \epsilon_a) \subseteq U_a$ for some ϵ_a . Now let $\epsilon'_a = \min(\epsilon_a, \min\{|a - b| * 0.25 \mid \forall b \in B\})$. Thus $[a, a + \epsilon'_a) \subseteq [a, a + \epsilon_a)$. Do the same $\forall b \in B$ instead of ϵ use δ . Let $U = \bigcup_{a \in A} [a, a + \epsilon'_a)$ and $V = \bigcup_{b \in B} [b, b + \delta'_b)$. $\forall x \in U \cap V \implies \exists a \in A, b \in B \ni x - a < \epsilon'_a$ and $x - b < \delta'_b$. However this is a contradiction. Thus $\mathbb{R}_{LL}$ is normal $\square$

Problem 4.6a. Consider $\mathbb{R}^2$ with the standard topology. Let $p \in \mathbb{R}^2 \ni p \notin A$, A a closed set. Show $\inf \{d(a, p) \mid a \in A\} > 0$

Solution. By definition $\forall x, y \in \mathbb{R}^2, d(x, y) \geq 0$. Assume the contrary, $\inf \{d(a, p) \mid a \in A\} = 0$. Thus $\forall r > 0, \exists a \in A \ni d(p, a) < r \implies a \in B(p, r) \cap A$. Thus $\forall U \in \mathcal{T} \ni p \in U, \exists \epsilon > 0 \ni B(p, \epsilon) \subseteq U \implies (U \setminus p) \cap A \neq \emptyset \implies p$ is a limit point of A which implies $p \in A$ which is a contradiction of the fact that $p \notin A$. $\blacksquare$

Problem 4.6b. Show that $\mathbb{R}^2$ with standard topology is regular

Proof. $\forall p \in \mathbb{R}^2$ and for all closed sets $A \subset \mathbb{R}^2 \ni p \notin A$. By the above $\delta := \inf \{d(a, p) \mid a \in A\} * 0.25 > 0$. Let $U = B(p, \delta)$. Let $V = \bigcup_{a \in A} B(a, \delta)$. $p \in U$ and $A \in V$.

Assume the contrary that $\exists x \in U \cap V$. Thus $d(x, p) < \delta$ and $\exists a \in A \ni d(x, a) < \delta$. Thus $d(x, a) \leq d(x, p) + d(p, a) < \delta + \delta = 2\delta$. However $d(x, a) \geq 4\delta$ which is a contradiction. Thus $U \cap V = \emptyset$. $\square$

Problem 4.6c. Find two disjoint closed subsets A and B of $\mathbb{R}^2$ with the standard topology such that

$$\inf \{d(a, b) \mid a \in A, b \in B\} = 0$$

Proof. Let $A = \mathbb{N} \times \{0\}$ and let $B = \{n + \frac{1}{n} \mid n \in \mathbb{N}\} \times \{0\}$. $\square$

Claim 4.6d. Show $\mathbb{R}^2$ with standard topology is normal

Proof. For each point a take the minimum distance $\square$

Claim 4.7a. A $\mathcal{T}_2$ -spaces (Hausdorff) is a $\mathcal{T}_1$ -space

Proof. Let $(X, \mathcal{T})$ be a $\mathcal{T}_2$ space. $\forall x, y \in X, \exists U, V \in \mathcal{T} \ni x \in U$ and $y \in V$ such that $U \cap V = \emptyset$. Thus $x \notin V$ and $y \notin U$. Thus $(X, \mathcal{T})$ is a $\mathcal{T}_1$ space. $\square$

Claim 4.7b. A $\mathcal{T}_3$ -space is a Hausdorff space

Proof. Since X is $\mathcal{T}_1, \forall x, y \in X \ni x \neq y \exists A \in \mathcal{T} \ni x \in A, y \notin A$. Thus $y_1 \in A$ which is closed. Since X is regular, there exists $U, V \in \mathcal{T} \ni x \in U, A \subseteq V, U \cap V = \emptyset$. Thus $y \in V$ and $y \notin U$. Thus $x \in U, y \in V, U \cap V = \emptyset$. Thus X is a Hausdorff space. $\square$

Claim 4.7c. A $\mathcal{T}_4$ space (normal and $\mathcal{T}_1$) is a $\mathcal{T}_3$ space (regular and $\mathcal{T}_1$)

Proof. $\forall x \in X$ and for all closed subsets $A \subset X$ such that $x \notin A$. Since X is $\mathcal{T}_4$ it is also $\mathcal{T}_1$. Thus $\forall y \in X$, there $\exists V_y \in \mathcal{T} \ni y \in V_y, x \notin V_y$. Thus $\forall y \in X \ni y \neq x, y \in \bigcup_{y \in X \setminus \{x\}} V_y$ but $x \notin \bigcup_{y \in X \setminus \{x\}} V_y$. Thus $(\bigcup_{y \in X \setminus \{x\}} V_y)^c = \{x\}$ is closed. Thus since $x \notin A \implies \{x\} \cap A = \emptyset$. Since X is $\mathcal{T}_4$, that means $\exists U, V \in \mathcal{T} \ni x \in U, A \subseteq V, U \cap V = \emptyset$. Thus X is regular and thus X is $\mathcal{T}_3$ $\square$

Claim 4.8. *A topological space X is regular iff $\forall p \in X$ and $\forall U \in \mathcal{T} \ni p \in U$ there is a open set V containing p such that $\overline{V} \subseteq U$*

Proof. $\implies$: Suppose X is a regular space. $\forall p \in X$ and $\forall U \in \mathcal{T} \ni p \in U$, $p \notin U^c$, a closed set. By definition of regular, $\exists V, W \in \mathcal{T} \ni p \in V, U^c \subseteq W, V \cap W = \emptyset$. Thus $p \in V \subseteq W^c$ and $W^c \subseteq U$. Let $W' = \text{Int}(W^c)$, $p \in V \subseteq W'$ and $\overline{W'} \subseteq W^c \subseteq U$.

$\Leftarrow$: Suppose that $\forall p \in X$ and $\forall U \in \mathcal{T} \ni p \in U$, $\exists V \in \mathcal{T} \ni p \in V, \overline{V} \subseteq U$. $\forall A \in \mathcal{C}$, closed subsets, such that $p \notin A$. A^c is a open subset. Thus $\exists V_1 \in \mathcal{T} \ni \overline{V_1} \subseteq A^c$. Thus $A \subseteq X - \overline{V_1}$. Since $V_1 \subset V_1 \implies V_1 \cap (X - \overline{V_1}) = \emptyset$. $p \in V_1$ and $A \subseteq X - \overline{V_1}$. Thus X is regular. $\square$

Claim 4.9. *A topological space X is regular iff for all closed sets A and $\forall U \in \mathcal{T} \ni A \subseteq U$ there is a open set V containing A such that $\overline{V} \subseteq U$*

Proof. $\implies$: Suppose X is a normal space. $\forall A \subseteq X$, closed, and $\forall U \in \mathcal{T} \ni A \subseteq U$. $A \not\subseteq U^c$. By the definition of regular, $\exists V, W \in \mathcal{T} \ni A \subseteq V, U^c \subseteq W, V \cap W = \emptyset$. Thus $A \subseteq V \subseteq W^c \subseteq U$. Let $W' = \text{Int}(W^c)$. Thus $A \subseteq V \subseteq W'$ and $\overline{W'} \subseteq W^c \subseteq U$.

$\Leftarrow$: Suppose for all closed sets A and $\forall U \in \mathcal{T} \ni A \subseteq U$ there is a open set V containing A such that $\overline{V} \subseteq U$. $\forall D \in \mathcal{C} \ni A \cap D = \emptyset$. D^c is a open subset. Thus $\exists U \in \mathcal{T} \ni A \subseteq U \subseteq \overline{U} \subseteq D^c$. Thus $D \subseteq X \setminus \overline{U}$ and $U \cap (X \setminus \overline{U}) = \emptyset$. Thus X is normal. $\square$

Claim 4.10. *A topological space X is normal iff for each pair of disjoint closed sets A and B , there are disjoint open sets U and V such that $A \subseteq U$ and $B \subseteq V$ and $\overline{U} \cap \overline{V} = \emptyset$*

Proof. $\Leftarrow$: Trivial because if closure is disjoint then set is disjoint.

$\implies$: Suppose X is normal. Then for any two disjoint closed sets A and B , $\exists U', V' \in \mathcal{T} \ni A \subseteq U', B \subseteq V', U' \cap V' = \emptyset$. By claim 4.9, $\exists U \in \mathcal{T} \ni A \subseteq U \subseteq \overline{U} \subseteq U'$ and similarly for $B \subseteq V$. Since $U' \cap V' = \emptyset$, thus $\overline{U} \cap \overline{V} = \emptyset$. Thus forward direction is true. $\square$

Claim 4.11. (Shrinking Lemma) *A topological space X is normal iff $\forall U, V \in \mathcal{T} \ni U \cup V = X$ there exists open sets U', V' such that $\overline{U'} \subseteq U$ and $\overline{V'} \subseteq V$ and $U' \cup V' = X$*

Proof. $\implies$: Suppose X is normal. Then $\forall U, V \in \mathcal{T} \ni U \cup V = X \implies U^c \cap V^c = \emptyset$. By normality and 4.9, $\exists M, N \in \mathcal{T} \ni U^c \subseteq M, V^c \subseteq N, \overline{M} \cap \overline{N} = \emptyset$. Thus $U^c \subseteq M \subseteq \overline{M}$ and $V^c \subseteq N \subseteq \overline{N}$. Thus $X \setminus \overline{M} \subseteq X \setminus \overline{N} \subseteq U$ and similarly $X - \text{cl}(N) \subseteq X - N \subseteq V$. $X - N$ and $X - M$ is closed thus $\text{cl}(X - \text{cl}(N)) \subseteq X - N$ and similarly $\text{cl}(X - \text{cl}(M)) \subseteq X - M$ and $X - \text{cl}(M), X - \text{cl}(N) \in \mathcal{T}$. Since $\text{cl}(M) \cap \text{cl}(N) = \emptyset \implies X - \text{cl}(M) \cup X - \text{cl}(N) = X$.

$\Leftarrow$: Suppose the latter half of the theorem is true. $\forall A, B \in \mathcal{C} \ni A \cap B = \emptyset \implies A^c \cup B^c = X$. Thus $\exists U', V' \in \mathcal{T} \ni \text{cl}(U') \subseteq A^c$ and $\text{cl}(V') \subseteq B^c$ such that $U' \cup V' = X \implies \text{cl}(U') \cup \text{cl}(V') = X$. Thus $A \subseteq X - \text{cl}(U')$, $B \subseteq X - \text{cl}(V')$ and $X - \text{cl}(U') \cap X - \text{cl}(V') = \emptyset$ $\square$

Claim 4.15. *Order Topologies are T_1 , Hausdorff, regular, and normal*

Proof. Let X be a topological space with a order topology. $\forall x, y \in X$, wlog $x < y$

1. Suppose $\exists b \in X \ni x < b < y$. Then $U := \{a \mid a < b\}$ and $V := \{a \mid a > b\}$. Thus $x \in U$ and $y \in V$ and $U \cap V = \emptyset$.
2. Otherwise. Let $U := \{a \mid a < y\}$ and $V := \{a \mid a > x\}$. $U \cap V = \emptyset$, because otherwise it reduces to the case above. Moreover $x \in U$ and $y \in V$.

Thus X is Hausdorff.

It suffices to prove that X is T_4 ie normal.

Define the following functions $\forall A \subseteq X$,

$f_A := A \rightarrow X \cup \{-\infty\}$ where $a \rightarrow \max(\{y \mid y \in X \setminus A, y < a\} \cup \{-\infty\})$

$g_A := A \rightarrow X \cup \{\infty\}$ where $a \rightarrow \min(\{y \mid y \in X \setminus A, y > a\} \cup \{\infty\})$.

Also define the intervals, $\forall a, b \in A \ni a < b$ $(-\infty, a) := \{x \mid x < a\}$, $(a, b) := \{x \mid a < x < b\}$, $(a, \infty) := \{x \mid x > a\}$.

$\forall A, B \in \mathcal{C} \ni A \cap B = \emptyset$. Define $\forall a \in A, \forall b \in B$ $U_a := (f_A(a), g_A(a))$, $V_b := (f_B(b), g_B(b))$. Thus $a \in U_a$ and $b \in V_b$. $\forall x \in U_a \cap V_b \implies f_A(a) < x < g_A(a)$ and $f_B(b) < x < g_B(b)$ which means $x \in A \cap B$ which is a contradiction. Thus $U_a \cap V_b = \emptyset$. Let $U := \bigcup_{a \in A} U_a$ and $V := \bigcup_{b \in B} V_b$. $U \cap V = \emptyset$ by above. But $A \subseteq U$ and $B \subseteq V$. Thus X is normal thus T_4 thus also regular. $\square$

Seperation Properties and Products

Claim 4.16. *Let X and Y be Hausdorff. Then $X \times Y$ is Hausdorff.*

Proof. $\forall (x_1, y_1), (x_2, y_2) \in X \times Y \ni x_1 \neq x_2 \text{ or } y_1 \neq y_2$.

1. If $x_1 \neq x_2$, then since X is Hausdorff $\exists U_X, V_X \in \mathcal{T}_X \ni x_1 \in U_X, x_2 \in V_X, U_X \cap V_X = \emptyset$. $Y \in \mathcal{T}_Y$ so let $U = U_X \times Y$ and $V = V_X \times Y$. Since $U_X \cap V_X = \emptyset \implies U \cap V = \emptyset$ and since $x_1 \in U_X \implies (x_1, y_1) \in U_X \times Y$ and similarly $(x_2, y_2) \in V$
2. If $y_1 \neq y_2$, since Y is Hausdorff $\exists U_Y, V_Y \in \mathcal{T}_Y \ni y_1 \in U_Y, y_2 \in V_Y, U_Y \cap V_Y = \emptyset$. Let $U = X \times U_Y$ and let $V = X \times V_Y$

Thus $X \times Y$ is Hausdorff. □

Claim 4.17. *Let X and Y be regular. Then $X \times Y$ is regular*

Proof. $\forall p \in X \times Y$ and $\forall A \in \mathcal{T} \ni p \in A$, by definition of product space $\exists \{S_\alpha\}_{\alpha \in \lambda} \subset \mathcal{T}_X, \exists \{T_\alpha\}_{\alpha \in \lambda} \in \mathcal{T}_Y \ni A = \bigcup_{\alpha \in \lambda} S_\alpha \times T_\alpha$ thus $\exists \alpha \in \lambda \ni p \in S_\alpha \times T_\alpha$. Let $p = (x, y)$ for $x \in X$ and $y \in Y$. Thus $x \in S_\alpha$ and $y \in T_\alpha$. Since X and Y are regular by 4.8, $\exists U \in \mathcal{T}_X \ni x \in U \subseteq \bar{U} \subseteq S_\alpha$ and $\exists V \in \mathcal{T}_Y \ni y \in V \subseteq \bar{V} \subseteq T_\alpha$. Thus $p \in U \times V, \bar{U} \times \bar{V} \subseteq A$ and $\text{cl}(U) \times \text{cl}(V)$ is closed in $X \times Y$ and $\text{cl}(U \times V) \subseteq \text{cl}(U) \times \text{cl}(V)$ (actually equality). Thus by 4.8, $X \times Y$ is regular. □

1 Heredity

Claim 4.19. *Every Hausdorff space is heriditarilly Hausdorff*

Proof. Let X be a Hausdorff space. $\forall Y \subset X, \forall x, y \in Y \ni x \neq y$ since X is Hausdorff $\exists U, V \in \mathcal{T} \ni x \in U, y \in V, U \cap V = \emptyset$. Thus $x \in U \cap Y \in \mathcal{T}_Y$ and $y \in V \cap Y \in \mathcal{T}_Y$ and $U \cap Y \cap V \cap Y = \emptyset$. Thus Y is Hausdorff. Thus X is heriditarilly Hausdorff. $\square$

Claim 4.20. *Every regular space is heriditarilly regular*

Proof. Let X be a regular space. $\forall Y \subset X, \forall y \in Y$ and $\forall A \in \mathcal{C}_Y \ni y \notin A$ by Claim 3.28 $\exists B \in \mathcal{C}_X \ni B \cap Y = A \implies y \notin B$. Since X is regular, $\exists U, V \in \mathcal{T} \ni x \in U, B \subseteq V, U \cap V = \emptyset$. Thus $x \in U \cap Y \in \mathcal{T}_Y$ and $A \subseteq V \cap Y \in \mathcal{T}_Y$ and $(U \cap Y) \cap (V \cap Y) = \emptyset$. Thus Y is regular. Thus X is heriditarilly regular. $\square$

Claim 4.21. $2^{\mathbb{R}}$ is normal

Proof. $\forall C, D \in \mathcal{C} \ni C \cap D = \emptyset$. Let $\square$

Definition. The **Tychonoff Plank** is a product of two ordinal spaces $(\omega_0 + 1) \times (\omega_1 + 1)$ $\blacksquare$

Claim 4.22a. *The Tychonoff Plank is normal*

Proof. $\square$

Claim 4.23. *Let A be a closed subset of a normal space X . Then A is normal when given the subspace topology*

Proof. $\forall C, D \subseteq \mathcal{C}_A \ni C \cap D = \emptyset$, by claim 3.29 $\text{cl}_X(C) \cap A = C$ and $\text{cl}_X(D) \cap A = D$. Assume the contrary $\exists x \in \text{cl}_X(C) \cap \text{cl}_X(D)$, thus $\forall U \in \mathcal{T}_X \ni x \in U, (U - \{x\}) \cap C \cap D \neq \emptyset \implies (U - \{x\}) \cap A \neq \emptyset$ since $C, D \subseteq A$. Thus $x \in A$, since A is closed. Thus $x \in C$ and $x \in D$ which is a contradiction since $C \cap D = \emptyset$. Thus $\text{cl}_X(C) \cap \text{cl}_X(D) = \emptyset$. By the normality of X , $\exists U, V \in \mathcal{T}_X \ni \text{cl}_X(C) \subseteq U, \text{cl}_X(D) \subseteq V, U \cap V = \emptyset$. Thus $C \subseteq (U \cap A) \in \mathcal{T}_A$ and $D \subseteq (V \cap A) \in \mathcal{T}_A$ and $(U \cap A) \cap (V \cap A) = \emptyset$. Thus $(A, \mathcal{T}_A)$ is normal. $\square$

Problem 4.25. *Let Y be a subspace of X , and let $A, B \in \mathcal{C}_Y \ni A \cap B = \emptyset$, show $\text{cl}_X(A) \cap B = \emptyset$ and $\text{cl}_X(B) \cap A = \emptyset$*

Proof. Assume the contrary, $\exists x \in \text{cl}_X(A) \cap B \implies x \in B \implies x \in Y$. Thus $x \in \text{cl}_X(A) \implies x \in \text{cl}_X(A) \cap Y = A$, claim 3.29. Thus $x \in A \cap B = \emptyset$, a contradiction. Thus $\text{cl}_X(A) \cap B = \emptyset$.

Arguement is symmetric for $\text{cl}_X(B) \cap A = \emptyset$. $\square$

Problem 4.26. *The space X is a completely normal space if and only if X is hereditarily normal.*

Proof. $\implies$: Suppose X is completely normal. $\forall Y \subseteq X, \forall A, B \in \mathcal{C}_Y \ni A \cap B = \emptyset$. By problem 4.25, $\text{cl}_X(A) \cap B = \emptyset$ and $\text{cl}_X(B) \cap A = \emptyset$ thus A and B are separated sets in X . Since X is completely normal, $\exists U, V \in \mathcal{T}_X \ni A \subseteq U, B \subseteq V, U \cap V = \emptyset$. Thus $A \subseteq U \cap Y \in \mathcal{T}_Y$ and $B \subseteq V \cap Y \in \mathcal{T}_Y$ and $(U \cap Y) \cap (V \cap Y) = \emptyset$. Thus Y is normal. Thus X is hereditarily normal.

$\impliedby$: Suppose X is hereditarily normal. $\forall A, B \subseteq X \ni \text{cl}(A) \cap B = \emptyset, \text{cl}(B) \cap A = \emptyset$. Let $Y = X \setminus (\text{cl}(A) \cap \text{cl}(B))$ is open. Assume the contrary, $\exists x \in \text{cl}_Y(A) \cap \text{cl}_Y(B)$, then $x \in \text{cl}_Y(A) = \text{cl}(A) \cap Y$ and $x \in \text{cl}(B) \cap Y$ thus $x \in \text{cl}(A) \cap \text{cl}(B)$ which is a contradiction since $Y = X \setminus (\text{cl}(A) \cap \text{cl}(B))$ thus $x \notin \text{cl}(A) \cap \text{cl}(B)$. Thus by contradiction $\text{cl}_Y(A)$ is disjoint from $\text{cl}_Y(B)$. Thus since X is hereditarily normal, Y is normal. Thus $\exists U, V \in \mathcal{T}_Y \ni \text{cl}_Y(A) \subseteq U, \text{cl}_Y(B) \subseteq V, U \cap V = \emptyset$. Since $U = Y \cap U'$ and $V = Y \cap V'$ for $U', V' \in \mathcal{T}$, $U, V \in \mathcal{T}$ by closure of finite intersections. Thus $A \subseteq U$ and $B \subseteq V$ and $U \cap V = \emptyset$. Thus X is completely normal. $\square$

Problem 4.27.

Solution. 1. The subspace topology of $S = (0, 1) \cup \{2\}$ has the open set $\{2\} = (1.5, 2.5) \cap S$. You can't get the open set $\{2\}$ with a order topology because $\forall a \in S, \exists b \in S \ni a < b < 2$ because a must be in $(0, 1)$. Thus if $a \in (0, 1)$ take $b = \frac{1+a}{2}$ $a < b < 1 < 2$, thus $b \in (a, \infty)$.

2. Take $L := \{(x, \frac{1}{2}) \mid x \in [0, 1]\}$. $\mathcal{T}_L$ is the discrete topology. Thus $\{(\frac{1}{2}, \frac{1}{2})\}$ is a open set. $\forall a < \frac{1}{2}, b > \frac{1}{2}$, take $a' = \frac{a+0.5}{2}$ and $b' = \frac{b+0.5}{2}$ thus $(a', \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}) \in ((a, \frac{1}{2}), (1, \frac{1}{2}])$ and $(b', \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}) \in [(0, \frac{1}{2}), (b, \frac{1}{2}))$ and $\{a', b'\} \times \frac{1}{2} \subseteq (a \times \frac{1}{2}, b \times \frac{1}{2})$. Thus $\mathcal{T}_L$ is not a order topology

3. $L := \{(x, 1) \mid x \in [0, 1]\}$

■

Claim 4.28. *Order topologies are hereditarily normal*

Proof. $\forall Y \subseteq X$.

Let $Y \cup \{\infty\}$ have the ordering induced by Y and where $\forall y \in Y, \infty > y$. Similarly let $Y \cup \{-\infty\}$ have the ordering induced by Y where $\forall y \in Y, -\infty < y$.

Define the following functions $\forall A \subseteq Y$: $f_A : A \rightarrow Y \cup \{-\infty\}$ and $g_A : A \rightarrow Y \cup \{\infty\}$. Define f_A where $a \rightarrow \max(\{x \mid x \in Y \setminus A, x < a\} \cup \{-\infty\})$. Define g_A where $a \rightarrow \min(\{x \mid x \in Y \setminus A, x > a\} \cup \{\infty\})$.

Define the function $(,) := Y \times Y \rightarrow 2^Y$ where $\forall a, b \in Y, (a, b) \rightarrow \{x \in Y \mid a < x < b\}, (a, \infty) \rightarrow \{x \in Y \mid a < x\}, (-\infty, b) \rightarrow \{x \in Y \mid b < x\}$.

$\forall A, B \in \mathcal{C}_Y \ni A \cap B = \emptyset$. Define $\forall a \in A, \forall b \in B, U_a := (f_A(a), g_A(a)), V_b := (f_B(b), g_B(b))$. Thus $a \in U_a$ and $b \in V_b$.

$\forall (a, b) \in A \times B, \forall x \in U_a \cap V_b \implies f_A(a) < x < g_A(a)$ and $f_B(b) < x < g_B(b)$ which means $x \in A \cap B$ which is a contradiction. Thus $U_a \cap V_b = \emptyset$. Let $U := \bigcup_{a \in A} U_a$ and $V := \bigcup_{b \in B} V_b$. $U \cap V = \emptyset$ by the above. But $A \subseteq U$ and $B \subseteq V$ Thus Y is normal. Thus X is hereditarily normal. $\square$

COME BACK HERE!!!!

2 The normality Lemma

Claim 4.29. (Normality Lemma). Let A and B be subsets of a topological space X , and let $\{U_i\}_{i \in \mathbb{N}}$ and $\{V_i\}_{i \in \mathbb{N}}$ be two collections of open sets such that

1. $A \subseteq \bigcup_{i \in \mathbb{N}} U_i$
2. $B \subseteq \bigcup_{i \in \mathbb{N}} V_i$
3. $\forall i \in \mathbb{N}, \overline{U_i} \cap B = \emptyset$ and $\overline{V_i} \cap A = \emptyset$

then there exists $U, V \in \mathcal{T} \ni A \subseteq U, B \subseteq V, U \cap V = \emptyset$

Proof. Let A and B be subsets of a topological space X , and let $\{U_i\}_{i \in \mathbb{N}}$ and $\{V_i\}_{i \in \mathbb{N}}$ be two collections of open sets which satisfy conditions 1, 2, 3

□

Claim 4.30. If X is normal and $C = \bigcup_{i \in \mathbb{N}} K_i$ is the union of closed sets K_i in X then C is normal

Proof. $\forall A, B \in \mathcal{C}_C \ni A \cap B = \emptyset. \text{cl}_X(A) \cap C = A$ and $\text{cl}_X(B) \cap C = B$.

□

Claim 4.31. Suppose X is regular and countable. Then X is normal

Proof.

□